ENERGY

Energy is all around us—the secret power behind everything in our world, from a bouncing ball to an exploding star. Energy is what makes things happen. It is what gives objects the ability to move, to glow with heat and light, or to make sounds. The ultimate source of all energy on Earth is the sun. Without energy, there would be no life.

TYPES OF ENERGY

Energy exists in many different forms. They are all closely related and each one can change into other types.

Sound energy
When objects vibrate, they make particles in the air vibrate, sending energy waves traveling to our ears, which we hear as sounds.

Potential energy
This is stored energy. Climb something, and you store potential energy to jump, roll, or dive back down.

Heat energy
Hot things have more energy than cold ones, because the particles inside them jiggle around more quickly.

Mechanical energy
Also known as elastic energy, this is the potential stored in stretched objects, such as a taut bow.

Electrical energy
Electricity is energy carried by charged particles called electrons moving through wires.

Nuclear energy
Atoms are bound together by energy, which they release when they split apart in nuclear reactions.

Light energy
Light travels at high speed and in straight lines. Like radio waves and X-rays, it is a type of electromagnetic energy.

Chemical energy
Food, fuel, and batteries store energy within the chemical compounds they are made of, which is released by reactions.

Kinetic energy
Moving things have kinetic energy. The heavier and faster they are, the more kinetic energy they have.

Measuring energy

Scientists measure energy in joules (J). One joule is the energy transferred to an object by a force of 1 newton (N) over a distance of 1 meter (m), also known as 1 newton meter (Nm).

• Energy of the sun
The sun produces four hundred octillion joules of energy each second!

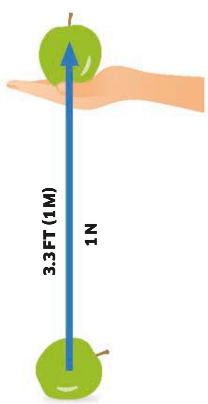
• Energy in water
To raise water temperature 1.8°F (1°C) takes 1 calorie (1/1,000 kilocalories).

• Energy in candles
A candle emits 80 J—or 80 W—of energy (mainly heat) each second.

• Energy in food
The energy released by food is measured in kilo-calories: 1 kcal is 4,184 J.

• Energy of a light bulb
An LED uses 15 watts (W), or 15 J, of electrical energy each second.

• Tiny amounts of energy
Ergs measure tiny units of energy. There are 10 million ergs in 1 J.



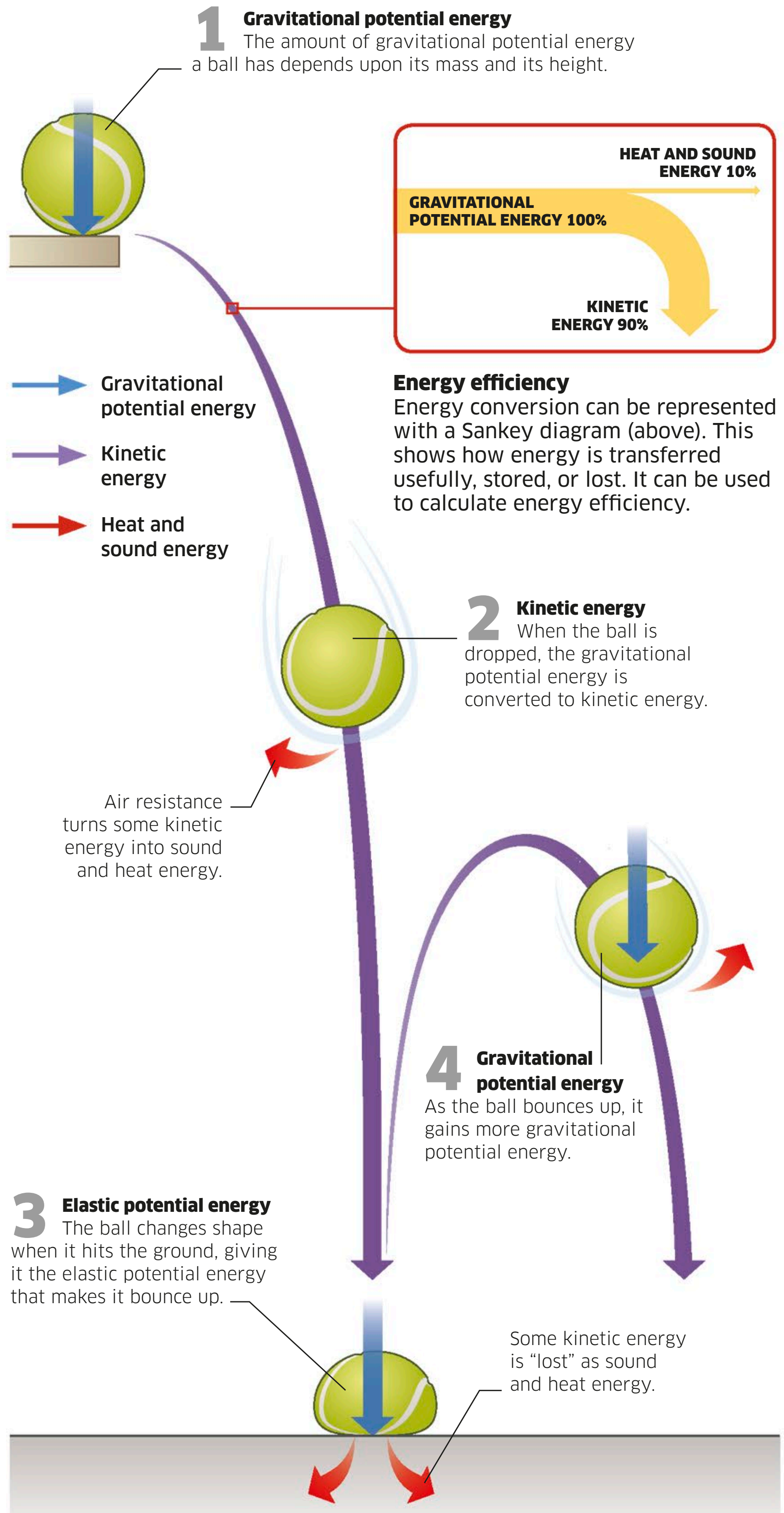
Lifting an apple
One joule is roughly equivalent to lifting an apple 3.3 ft (1 m).

CONSERVATION OF ENERGY

There's a fixed amount of energy in the universe that cannot be created or destroyed, but it can be transferred from one object to another and converted into different forms.

Energy conversion

The total amount of energy at the start of a process is always the same at the end, even though it has been converted into different forms. When you switch on a lamp, for example, most of the electrical energy is converted into light energy—but some will be lost as heat energy. However, the total amount of energy that exists always stays the same.



Energy efficiency
Energy conversion can be represented with a Sankey diagram (above). This shows how energy is transferred usefully, stored, or lost. It can be used to calculate energy efficiency.